100 Computer Vision MCQs

Complete English Practice & Interview Guide

Questions • 4 Options • Correct Answer • Explanation of Every Option

Topics Covered: Image Processing, CNNs, Transfer Learning, Object Detection, Segmentation, OpenCV, OCR, Face Detection, Feature Extraction, Camera Geometry, 3D Vision, Vision Transformers, Multimodal Vision, Evaluation Metrics, Deployment, Edge AI, and Computer Vision Fundamentals.

How to Use This Guide

Each question follows the same format: a computer vision question, four multiple-choice options, the correct answer, and an explanation of why the correct option is right and why each other option is incorrect. The questions are designed for university exams, interviews, AI/ML internships, and practical computer vision revision.

Difficulty: Beginner to intermediate, with practical and interview-oriented questions.

Question 1: What is the primary purpose of image classification?

A. To assign an image to one or more predefined categories

B. To generate a 3D model from an image

C. To compress an image

D. To remove all image noise

Correct Answer: A. To assign an image to one or more predefined categories

Why this is correct: To assign an image to one or more predefined categories directly matches the concept described in the question.

Why B is incorrect: To generate a 3D model from an image does not correctly describe the concept being tested in this question.

Why C is incorrect: To compress an image does not correctly describe the concept being tested in this question.

Why D is incorrect: To remove all image noise does not correctly describe the concept being tested in this question.

Question 2: Which color space is commonly used as the default representation for digital images?

A. RGB

B. XYZ only

C. CMYK only

D. YUV only

Correct Answer: A. RGB

Why this is correct: RGB directly matches the concept described in the question.

Why B is incorrect: XYZ only does not correctly describe the concept being tested in this question.

Why C is incorrect: CMYK only does not correctly describe the concept being tested in this question.

Why D is incorrect: YUV only does not correctly describe the concept being tested in this question.

Question 3: What does a pixel represent in a digital image?

A. A small unit containing image intensity/color information

B. A complete image

C. A neural network layer

D. A camera lens

Correct Answer: A. A small unit containing image intensity/color information

Why this is correct: A small unit containing image intensity/color information directly matches the concept described in the question.

Why B is incorrect: A complete image does not correctly describe the concept being tested in this question.

Why C is incorrect: A neural network layer does not correctly describe the concept being tested in this question.

Why D is incorrect: A camera lens does not correctly describe the concept being tested in this question.

Question 4: What is image segmentation?

A. Dividing an image into meaningful regions or pixels

B. Increasing image file size

C. Changing a PNG file to JPG

D. Converting an image to audio

Correct Answer: A. Dividing an image into meaningful regions or pixels

Why this is correct: Dividing an image into meaningful regions or pixels directly matches the concept described in the question.

Why B is incorrect: Increasing image file size does not correctly describe the concept being tested in this question.

Why C is incorrect: Changing a PNG file to JPG does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting an image to audio does not correctly describe the concept being tested in this question.

Question 5: Which task predicts a bounding box and class for objects in an image?

A. Object detection

B. Image compression

C. Image denoising

D. Image registration

Correct Answer: A. Object detection

Why this is correct: Object detection directly matches the concept described in the question.

Why B is incorrect: Image compression does not correctly describe the concept being tested in this question.

Why C is incorrect: Image denoising does not correctly describe the concept being tested in this question.

Why D is incorrect: Image registration does not correctly describe the concept being tested in this question.

Question 6: What is the main purpose of a convolutional neural network (CNN) in computer vision?

A. Learning spatial features from images

B. Storing images permanently

C. Replacing the camera sensor

D. Converting images directly into PDFs

Correct Answer: A. Learning spatial features from images

Why this is correct: Learning spatial features from images directly matches the concept described in the question.

Why B is incorrect: Storing images permanently does not correctly describe the concept being tested in this question.

Why C is incorrect: Replacing the camera sensor does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting images directly into PDFs does not correctly describe the concept being tested in this question.

Question 7: Which operation is fundamental to a convolutional layer?

A. Applying learnable filters across spatial locations

B. Sorting pixels alphabetically

C. Deleting all edges

D. Changing image metadata

Correct Answer: A. Applying learnable filters across spatial locations

Why this is correct: Applying learnable filters across spatial locations directly matches the concept described in the question.

Why B is incorrect: Sorting pixels alphabetically does not correctly describe the concept being tested in this question.

Why C is incorrect: Deleting all edges does not correctly describe the concept being tested in this question.

Why D is incorrect: Changing image metadata does not correctly describe the concept being tested in this question.

Question 8: What is pooling commonly used for in CNNs?

A. Reducing spatial dimensions and summarizing features

B. Increasing the number of image channels only

C. Adding random labels

D. Generating training annotations

Correct Answer: A. Reducing spatial dimensions and summarizing features

Why this is correct: Reducing spatial dimensions and summarizing features directly matches the concept described in the question.

Why B is incorrect: Increasing the number of image channels only does not correctly describe the concept being tested in this question.

Why C is incorrect: Adding random labels does not correctly describe the concept being tested in this question.

Why D is incorrect: Generating training annotations does not correctly describe the concept being tested in this question.

Question 9: Which pooling operation selects the largest value in a local region?

A. Max pooling

B. Min pooling

C. Average dropout

D. Median encoding

Correct Answer: A. Max pooling

Why this is correct: Max pooling directly matches the concept described in the question.

Why B is incorrect: Min pooling does not correctly describe the concept being tested in this question.

Why C is incorrect: Average dropout does not correctly describe the concept being tested in this question.

Why D is incorrect: Median encoding does not correctly describe the concept being tested in this question.

Question 10: What does stride mean in a convolutional layer?

A. The number of pixels the filter moves at each step

B. The number of classes in a dataset

C. The number of images in a batch

D. The number of channels in an RGB image

Correct Answer: A. The number of pixels the filter moves at each step

Why this is correct: The number of pixels the filter moves at each step directly matches the concept described in the question.

Why B is incorrect: The number of classes in a dataset does not correctly describe the concept being tested in this question.

Why C is incorrect: The number of images in a batch does not correctly describe the concept being tested in this question.

Why D is incorrect: The number of channels in an RGB image does not correctly describe the concept being tested in this question.

Question 11: What is padding in a convolutional neural network?

A. Adding values around an image before convolution

B. Removing labels from images

C. Encrypting the image

D. Increasing camera resolution

Correct Answer: A. Adding values around an image before convolution

Why this is correct: Adding values around an image before convolution directly matches the concept described in the question.

Why B is incorrect: Removing labels from images does not correctly describe the concept being tested in this question.

Why C is incorrect: Encrypting the image does not correctly describe the concept being tested in this question.

Why D is incorrect: Increasing camera resolution does not correctly describe the concept being tested in this question.

Question 12: What is the usual effect of 'same' padding with stride 1?

A. It preserves the spatial dimensions of the feature map

B. It always reduces the image to one pixel

C. It removes all channels

D. It doubles the image dimensions

Correct Answer: A. It preserves the spatial dimensions of the feature map

Why this is correct: It preserves the spatial dimensions of the feature map directly matches the concept described in the question.

Why B is incorrect: It always reduces the image to one pixel does not correctly describe the concept being tested in this question.

Why C is incorrect: It removes all channels does not correctly describe the concept being tested in this question.

Why D is incorrect: It doubles the image dimensions does not correctly describe the concept being tested in this question.

Question 13: Which activation function is widely used in CNN hidden layers?

A. ReLU

B. Softmax only

C. Linear only

D. JPEG

Correct Answer: A. ReLU

Why this is correct: ReLU directly matches the concept described in the question.

Why B is incorrect: Softmax only does not correctly describe the concept being tested in this question.

Why C is incorrect: Linear only does not correctly describe the concept being tested in this question.

Why D is incorrect: JPEG does not correctly describe the concept being tested in this question.

Question 14: What problem can ReLU help reduce compared with sigmoid in deep networks?

A. Vanishing gradients in many practical settings

B. Image file corruption

C. Camera blur

D. Class imbalance automatically

Correct Answer: A. Vanishing gradients in many practical settings

Why this is correct: Vanishing gradients in many practical settings directly matches the concept described in the question.

Why B is incorrect: Image file corruption does not correctly describe the concept being tested in this question.

Why C is incorrect: Camera blur does not correctly describe the concept being tested in this question.

Why D is incorrect: Class imbalance automatically does not correctly describe the concept being tested in this question.

Question 15: What is a feature map in a CNN?

A. The output produced by applying a filter to an input

B. A list of image filenames

C. A camera calibration file

D. A compressed JPEG

Correct Answer: A. The output produced by applying a filter to an input

Why this is correct: The output produced by applying a filter to an input directly matches the concept described in the question.

Why B is incorrect: A list of image filenames does not correctly describe the concept being tested in this question.

Why C is incorrect: A camera calibration file does not correctly describe the concept being tested in this question.

Why D is incorrect: A compressed JPEG does not correctly describe the concept being tested in this question.

Question 16: What is transfer learning in computer vision?

A. Using knowledge from a model trained on another dataset/task as a starting point

B. Transferring images between folders

C. Moving a model from CPU to GPU only

D. Copying labels into filenames

Correct Answer: A. Using knowledge from a model trained on another dataset/task as a starting point

Why this is correct: Using knowledge from a model trained on another dataset/task as a starting point directly matches the concept described in the question.

Why B is incorrect: Transferring images between folders does not correctly describe the concept being tested in this question.

Why C is incorrect: Moving a model from CPU to GPU only does not correctly describe the concept being tested in this question.

Why D is incorrect: Copying labels into filenames does not correctly describe the concept being tested in this question.

Question 17: Why is transfer learning useful when the target dataset is small?

A. A pretrained model can provide useful learned visual representations

B. It eliminates the need for any validation data

C. It guarantees 100% accuracy

D. It removes all preprocessing requirements

Correct Answer: A. A pretrained model can provide useful learned visual representations

Why this is correct: A pretrained model can provide useful learned visual representations directly matches the concept described in the question.

Why B is incorrect: It eliminates the need for any validation data does not correctly describe the concept being tested in this question.

Why C is incorrect: It guarantees 100% accuracy does not correctly describe the concept being tested in this question.

Why D is incorrect: It removes all preprocessing requirements does not correctly describe the concept being tested in this question.

Question 18: Which model is a classic CNN architecture introduced by Alex Krizhevsky and colleagues?

A. AlexNet

B. K-Means

C. SIFT

D. HarrisNet

Correct Answer: A. AlexNet

Why this is correct: AlexNet directly matches the concept described in the question.

Why B is incorrect: K-Means does not correctly describe the concept being tested in this question.

Why C is incorrect: SIFT does not correctly describe the concept being tested in this question.

Why D is incorrect: HarrisNet does not correctly describe the concept being tested in this question.

Question 19: Which architecture is known for using residual connections?

A. ResNet

B. Naive Bayes

C. KNN

D. SVM-R

Correct Answer: A. ResNet

Why this is correct: ResNet directly matches the concept described in the question.

Why B is incorrect: Naive Bayes does not correctly describe the concept being tested in this question.

Why C is incorrect: KNN does not correctly describe the concept being tested in this question.

Why D is incorrect: SVM-R does not correctly describe the concept being tested in this question.

Question 20: What is the key idea behind residual connections?

A. Learning a residual mapping that helps information and gradients flow through deep networks

B. Removing all convolutional layers

C. Using only grayscale images

D. Replacing training with manual labeling

Correct Answer: A. Learning a residual mapping that helps information and gradients flow through deep networks

Why this is correct: Learning a residual mapping that helps information and gradients flow through deep networks directly matches the concept described in the question.

Why B is incorrect: Removing all convolutional layers does not correctly describe the concept being tested in this question.

Why C is incorrect: Using only grayscale images does not correctly describe the concept being tested in this question.

Why D is incorrect: Replacing training with manual labeling does not correctly describe the concept being tested in this question.

Question 21: Which architecture introduced the Inception module?

A. GoogLeNet/Inception

B. ResNet

C. LeNet-1

D. YOLO only

Correct Answer: A. GoogLeNet/Inception

Why this is correct: GoogLeNet/Inception directly matches the concept described in the question.

Why B is incorrect: ResNet does not correctly describe the concept being tested in this question.

Why C is incorrect: LeNet-1 does not correctly describe the concept being tested in this question.

Why D is incorrect: YOLO only does not correctly describe the concept being tested in this question.

Question 22: Which CNN architecture is known for a simple stack of small 3×3 convolutions?

A. VGG

B. GAN

C. R-CNN

D. SIFT

Correct Answer: A. VGG

Why this is correct: VGG directly matches the concept described in the question.

Why B is incorrect: GAN does not correctly describe the concept being tested in this question.

Why C is incorrect: R-CNN does not correctly describe the concept being tested in this question.

Why D is incorrect: SIFT does not correctly describe the concept being tested in this question.

Question 23: What is image augmentation?

A. Creating varied training examples through transformations of existing images

B. Deleting duplicate classes

C. Changing a model's optimizer after deployment

D. Converting RGB to a database

Correct Answer: A. Creating varied training examples through transformations of existing images

Why this is correct: Creating varied training examples through transformations of existing images directly matches the concept described in the question.

Why B is incorrect: Deleting duplicate classes does not correctly describe the concept being tested in this question.

Why C is incorrect: Changing a model's optimizer after deployment does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting RGB to a database does not correctly describe the concept being tested in this question.

Question 24: Which is a common image augmentation technique?

A. Horizontal flipping

B. Changing a label to a random class

C. Deleting all pixels

D. Removing the dataset

Correct Answer: A. Horizontal flipping

Why this is correct: Horizontal flipping directly matches the concept described in the question.

Why B is incorrect: Changing a label to a random class does not correctly describe the concept being tested in this question.

Why C is incorrect: Deleting all pixels does not correctly describe the concept being tested in this question.

Why D is incorrect: Removing the dataset does not correctly describe the concept being tested in this question.

Question 25: Why should labels be updated when an augmentation changes object geometry?

A. Because the object's location or orientation may no longer match the original annotation

B. Because augmentation always changes the class

C. Because RGB values become invalid

D. Because CNNs cannot use augmented images

Correct Answer: A. Because the object's location or orientation may no longer match the original annotation

Why this is correct: Because the object's location or orientation may no longer match the original annotation directly matches the concept described in the question.

Why B is incorrect: Because augmentation always changes the class does not correctly describe the concept being tested in this question.

Why C is incorrect: Because RGB values become invalid does not correctly describe the concept being tested in this question.

Why D is incorrect: Because CNNs cannot use augmented images does not correctly describe the concept being tested in this question.

Question 26: What is normalization in image preprocessing?

A. Scaling pixel values according to a defined numerical scheme

B. Adding object labels

C. Removing all edges

D. Converting every image to a video

Correct Answer: A. Scaling pixel values according to a defined numerical scheme

Why this is correct: Scaling pixel values according to a defined numerical scheme directly matches the concept described in the question.

Why B is incorrect: Adding object labels does not correctly describe the concept being tested in this question.

Why C is incorrect: Removing all edges does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting every image to a video does not correctly describe the concept being tested in this question.

Question 27: What is the common 8-bit range for a grayscale pixel?

A. 0 to 255

B. 0 to 1,000

C. -255 to 255 only

D. 1 to 10

Correct Answer: A. 0 to 255

Why this is correct: 0 to 255 directly matches the concept described in the question.

Why B is incorrect: 0 to 1,000 does not correctly describe the concept being tested in this question.

Why C is incorrect: -255 to 255 only does not correctly describe the concept being tested in this question.

Why D is incorrect: 1 to 10 does not correctly describe the concept being tested in this question.

Question 28: What is edge detection designed to identify?

A. Locations of strong intensity changes

B. Image file names

C. Audio frequencies

D. Database relationships

Correct Answer: A. Locations of strong intensity changes

Why this is correct: Locations of strong intensity changes directly matches the concept described in the question.

Why B is incorrect: Image file names does not correctly describe the concept being tested in this question.

Why C is incorrect: Audio frequencies does not correctly describe the concept being tested in this question.

Why D is incorrect: Database relationships does not correctly describe the concept being tested in this question.

Question 29: Which operator is commonly used for edge detection?

A. Sobel

B. Softmax

C. Adam

D. Dropout

Correct Answer: A. Sobel

Why this is correct: Sobel directly matches the concept described in the question.

Why B is incorrect: Softmax does not correctly describe the concept being tested in this question.

Why C is incorrect: Adam does not correctly describe the concept being tested in this question.

Why D is incorrect: Dropout does not correctly describe the concept being tested in this question.

Question 30: Which edge detector is known for using Gaussian smoothing followed by gradient and hysteresis steps?

A. Canny

B. K-Means

C. HOG

D. ORB

Correct Answer: A. Canny

Why this is correct: Canny directly matches the concept described in the question.

Why B is incorrect: K-Means does not correctly describe the concept being tested in this question.

Why C is incorrect: HOG does not correctly describe the concept being tested in this question.

Why D is incorrect: ORB does not correctly describe the concept being tested in this question.

Question 31: What is thresholding in image processing?

A. Separating pixels based on intensity or another criterion

B. Training a CNN without data

C. Increasing network depth

D. Changing a PNG into a video

Correct Answer: A. Separating pixels based on intensity or another criterion

Why this is correct: Separating pixels based on intensity or another criterion directly matches the concept described in the question.

Why B is incorrect: Training a CNN without data does not correctly describe the concept being tested in this question.

Why C is incorrect: Increasing network depth does not correctly describe the concept being tested in this question.

Why D is incorrect: Changing a PNG into a video does not correctly describe the concept being tested in this question.

Question 32: What is binary thresholding?

A. Converting pixels into two classes based on a threshold

B. Converting RGB into three classes

C. Adding two cameras

D. Creating two neural networks

Correct Answer: A. Converting pixels into two classes based on a threshold

Why this is correct: Converting pixels into two classes based on a threshold directly matches the concept described in the question.

Why B is incorrect: Converting RGB into three classes does not correctly describe the concept being tested in this question.

Why C is incorrect: Adding two cameras does not correctly describe the concept being tested in this question.

Why D is incorrect: Creating two neural networks does not correctly describe the concept being tested in this question.

Question 33: What is adaptive thresholding useful for?

A. Images with uneven illumination

B. Only perfectly uniform backgrounds

C. Increasing GPU memory

D. Generating 3D meshes

Correct Answer: A. Images with uneven illumination

Why this is correct: Images with uneven illumination directly matches the concept described in the question.

Why B is incorrect: Only perfectly uniform backgrounds does not correctly describe the concept being tested in this question.

Why C is incorrect: Increasing GPU memory does not correctly describe the concept being tested in this question.

Why D is incorrect: Generating 3D meshes does not correctly describe the concept being tested in this question.

Question 34: What is morphological erosion?

A. A morphological operation that can shrink foreground regions

B. A color-space conversion

C. A neural-network optimizer

D. An object detector

Correct Answer: A. A morphological operation that can shrink foreground regions

Why this is correct: A morphological operation that can shrink foreground regions directly matches the concept described in the question.

Why B is incorrect: A color-space conversion does not correctly describe the concept being tested in this question.

Why C is incorrect: A neural-network optimizer does not correctly describe the concept being tested in this question.

Why D is incorrect: An object detector does not correctly describe the concept being tested in this question.

Question 35: What is morphological dilation?

A. A morphological operation that can expand foreground regions

B. Reducing image resolution

C. Removing labels

D. Changing RGB to HSV

Correct Answer: A. A morphological operation that can expand foreground regions

Why this is correct: A morphological operation that can expand foreground regions directly matches the concept described in the question.

Why B is incorrect: Reducing image resolution does not correctly describe the concept being tested in this question.

Why C is incorrect: Removing labels does not correctly describe the concept being tested in this question.

Why D is incorrect: Changing RGB to HSV does not correctly describe the concept being tested in this question.

Question 36: What is the main purpose of an opening operation in morphology?

A. Erosion followed by dilation, often useful for removing small objects/noise

B. Dilation followed by convolution only

C. Color correction followed by resizing

D. Object detection followed by OCR

Correct Answer: A. Erosion followed by dilation, often useful for removing small objects/noise

Why this is correct: Erosion followed by dilation, often useful for removing small objects/noise directly matches the concept described in the question.

Why B is incorrect: Dilation followed by convolution only does not correctly describe the concept being tested in this question.

Why C is incorrect: Color correction followed by resizing does not correctly describe the concept being tested in this question.

Why D is incorrect: Object detection followed by OCR does not correctly describe the concept being tested in this question.

Question 37: What is the main purpose of a closing operation in morphology?

A. Dilation followed by erosion, often useful for filling small gaps

B. Erosion followed by image deletion

C. Cropping followed by JPEG compression

D. Classification followed by tracking

Correct Answer: A. Dilation followed by erosion, often useful for filling small gaps

Why this is correct: Dilation followed by erosion, often useful for filling small gaps directly matches the concept described in the question.

Why B is incorrect: Erosion followed by image deletion does not correctly describe the concept being tested in this question.

Why C is incorrect: Cropping followed by JPEG compression does not correctly describe the concept being tested in this question.

Why D is incorrect: Classification followed by tracking does not correctly describe the concept being tested in this question.

Question 38: What does image resizing change?

A. The spatial dimensions of an image

B. The semantic class automatically

C. The camera sensor

D. The original ground-truth meaning

Correct Answer: A. The spatial dimensions of an image

Why this is correct: The spatial dimensions of an image directly matches the concept described in the question.

Why B is incorrect: The semantic class automatically does not correctly describe the concept being tested in this question.

Why C is incorrect: The camera sensor does not correctly describe the concept being tested in this question.

Why D is incorrect: The original ground-truth meaning does not correctly describe the concept being tested in this question.

Question 39: What is interpolation used for during image resizing?

A. Estimating pixel values at new locations

B. Generating object labels

C. Training the optimizer

D. Detecting faces directly

Correct Answer: A. Estimating pixel values at new locations

Why this is correct: Estimating pixel values at new locations directly matches the concept described in the question.

Why B is incorrect: Generating object labels does not correctly describe the concept being tested in this question.

Why C is incorrect: Training the optimizer does not correctly describe the concept being tested in this question.

Why D is incorrect: Detecting faces directly does not correctly describe the concept being tested in this question.

Question 40: Which interpolation method is often fast and simple for image resizing?

A. Nearest neighbor

B. Backpropagation

C. Softmax

D. Batch normalization

Correct Answer: A. Nearest neighbor

Why this is correct: Nearest neighbor directly matches the concept described in the question.

Why B is incorrect: Backpropagation does not correctly describe the concept being tested in this question.

Why C is incorrect: Softmax does not correctly describe the concept being tested in this question.

Why D is incorrect: Batch normalization does not correctly describe the concept being tested in this question.

Question 41: What is object detection?

A. Finding and classifying objects along with their locations

B. Assigning one label to the entire image only

C. Removing image backgrounds

D. Converting images into text only

Correct Answer: A. Finding and classifying objects along with their locations

Why this is correct: Finding and classifying objects along with their locations directly matches the concept described in the question.

Why B is incorrect: Assigning one label to the entire image only does not correctly describe the concept being tested in this question.

Why C is incorrect: Removing image backgrounds does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting images into text only does not correctly describe the concept being tested in this question.

Question 42: Which output is commonly used to represent an object's location in 2D object detection?

A. A bounding box

B. A confusion matrix

C. A histogram only

D. A database key

Correct Answer: A. A bounding box

Why this is correct: A bounding box directly matches the concept described in the question.

Why B is incorrect: A confusion matrix does not correctly describe the concept being tested in this question.

Why C is incorrect: A histogram only does not correctly describe the concept being tested in this question.

Why D is incorrect: A database key does not correctly describe the concept being tested in this question.

Question 43: What does IoU stand for in object detection?

A. Intersection over Union

B. Image Output Utility

C. Input Object Unit

D. Inference Optimization Utility

Correct Answer: A. Intersection over Union

Why this is correct: Intersection over Union directly matches the concept described in the question.

Why B is incorrect: Image Output Utility does not correctly describe the concept being tested in this question.

Why C is incorrect: Input Object Unit does not correctly describe the concept being tested in this question.

Why D is incorrect: Inference Optimization Utility does not correctly describe the concept being tested in this question.

Question 44: What does IoU measure?

A. The overlap between two regions such as predicted and ground-truth boxes

B. The number of pixels in an image only

C. The speed of a camera

D. The number of model parameters

Correct Answer: A. The overlap between two regions such as predicted and ground-truth boxes

Why this is correct: The overlap between two regions such as predicted and ground-truth boxes directly matches the concept described in the question.

Why B is incorrect: The number of pixels in an image only does not correctly describe the concept being tested in this question.

Why C is incorrect: The speed of a camera does not correctly describe the concept being tested in this question.

Why D is incorrect: The number of model parameters does not correctly describe the concept being tested in this question.

Question 45: What is Non-Maximum Suppression (NMS) used for?

A. Reducing duplicate overlapping detections

B. Increasing image resolution

C. Training a tokenizer

D. Converting images to grayscale

Correct Answer: A. Reducing duplicate overlapping detections

Why this is correct: Reducing duplicate overlapping detections directly matches the concept described in the question.

Why B is incorrect: Increasing image resolution does not correctly describe the concept being tested in this question.

Why C is incorrect: Training a tokenizer does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting images to grayscale does not correctly describe the concept being tested in this question.

Question 46: What is YOLO best known for?

A. Real-time object detection using a one-stage detection approach

B. Image compression

C. Optical character recognition only

D. Audio classification

Correct Answer: A. Real-time object detection using a one-stage detection approach

Why this is correct: Real-time object detection using a one-stage detection approach directly matches the concept described in the question.

Why B is incorrect: Image compression does not correctly describe the concept being tested in this question.

Why C is incorrect: Optical character recognition only does not correctly describe the concept being tested in this question.

Why D is incorrect: Audio classification does not correctly describe the concept being tested in this question.

Question 47: What does the name YOLO commonly expand to?

A. You Only Look Once

B. Your Object Learns Once

C. Yield Only Local Objects

D. You Optimize Learning Outputs

Correct Answer: A. You Only Look Once

Why this is correct: You Only Look Once directly matches the concept described in the question.

Why B is incorrect: Your Object Learns Once does not correctly describe the concept being tested in this question.

Why C is incorrect: Yield Only Local Objects does not correctly describe the concept being tested in this question.

Why D is incorrect: You Optimize Learning Outputs does not correctly describe the concept being tested in this question.

Question 48: What is a one-stage object detector?

A. A detector that predicts object locations and classes in a unified detection process

B. A detector that needs exactly one image in the dataset

C. A detector without any learned parameters

D. A detector that only works on grayscale images

Correct Answer: A. A detector that predicts object locations and classes in a unified detection process

Why this is correct: A detector that predicts object locations and classes in a unified detection process directly matches the concept described in the question.

Why B is incorrect: A detector that needs exactly one image in the dataset does not correctly describe the concept being tested in this question.

Why C is incorrect: A detector without any learned parameters does not correctly describe the concept being tested in this question.

Why D is incorrect: A detector that only works on grayscale images does not correctly describe the concept being tested in this question.

Question 49: What is the main idea of two-stage object detectors such as Faster R-CNN?

A. First generate/propose candidate regions, then classify/refine them

B. First compress the image, then delete it

C. Use two cameras instead of one

D. Train two unrelated classifiers

Correct Answer: A. First generate/propose candidate regions, then classify/refine them

Why this is correct: First generate/propose candidate regions, then classify/refine them directly matches the concept described in the question.

Why B is incorrect: First compress the image, then delete it does not correctly describe the concept being tested in this question.

Why C is incorrect: Use two cameras instead of one does not correctly describe the concept being tested in this question.

Why D is incorrect: Train two unrelated classifiers does not correctly describe the concept being tested in this question.

Question 50: What does R-CNN stand for?

A. Regions with Convolutional Neural Networks

B. Random Convolutional Neural Network

C. Recursive Camera Neural Network

D. Region Color Normalization Network

Correct Answer: A. Regions with Convolutional Neural Networks

Why this is correct: Regions with Convolutional Neural Networks directly matches the concept described in the question.

Why B is incorrect: Random Convolutional Neural Network does not correctly describe the concept being tested in this question.

Why C is incorrect: Recursive Camera Neural Network does not correctly describe the concept being tested in this question.

Why D is incorrect: Region Color Normalization Network does not correctly describe the concept being tested in this question.

Question 51: What is instance segmentation?

A. Assigning a separate pixel mask to each detected object instance

B. Assigning one class to the whole image

C. Detecting only image edges

D. Compressing each object separately

Correct Answer: A. Assigning a separate pixel mask to each detected object instance

Why this is correct: Assigning a separate pixel mask to each detected object instance directly matches the concept described in the question.

Why B is incorrect: Assigning one class to the whole image does not correctly describe the concept being tested in this question.

Why C is incorrect: Detecting only image edges does not correctly describe the concept being tested in this question.

Why D is incorrect: Compressing each object separately does not correctly describe the concept being tested in this question.

Question 52: What is semantic segmentation?

A. Assigning a class label to each pixel without necessarily distinguishing individual instances of the same class

B. Drawing only bounding boxes

C. Detecting image metadata

D. Classifying only the whole image

Correct Answer: A. Assigning a class label to each pixel without necessarily distinguishing individual instances of the same class

Why this is correct: Assigning a class label to each pixel without necessarily distinguishing individual instances of the same class directly matches the concept described in the question.

Why B is incorrect: Drawing only bounding boxes does not correctly describe the concept being tested in this question.

Why C is incorrect: Detecting image metadata does not correctly describe the concept being tested in this question.

Why D is incorrect: Classifying only the whole image does not correctly describe the concept being tested in this question.

Question 53: What is panoptic segmentation?

A. A segmentation framework that combines semantic and instance-level understanding

B. A method for image compression

C. A type of camera lens

D. A grayscale conversion technique

Correct Answer: A. A segmentation framework that combines semantic and instance-level understanding

Why this is correct: A segmentation framework that combines semantic and instance-level understanding directly matches the concept described in the question.

Why B is incorrect: A method for image compression does not correctly describe the concept being tested in this question.

Why C is incorrect: A type of camera lens does not correctly describe the concept being tested in this question.

Why D is incorrect: A grayscale conversion technique does not correctly describe the concept being tested in this question.

Question 54: Which architecture is strongly associated with encoder-decoder segmentation?

A. U-Net

B. K-Means

C. SVM

D. AlexNet only

Correct Answer: A. U-Net

Why this is correct: U-Net directly matches the concept described in the question.

Why B is incorrect: K-Means does not correctly describe the concept being tested in this question.

Why C is incorrect: SVM does not correctly describe the concept being tested in this question.

Why D is incorrect: AlexNet only does not correctly describe the concept being tested in this question.

Question 55: Why is U-Net popular for biomedical image segmentation?

A. Its skip connections help preserve fine spatial information

B. It requires no training data

C. It works only with text

D. It cannot output masks

Correct Answer: A. Its skip connections help preserve fine spatial information

Why this is correct: Its skip connections help preserve fine spatial information directly matches the concept described in the question.

Why B is incorrect: It requires no training data does not correctly describe the concept being tested in this question.

Why C is incorrect: It works only with text does not correctly describe the concept being tested in this question.

Why D is incorrect: It cannot output masks does not correctly describe the concept being tested in this question.

Question 56: What is a segmentation mask?

A. An image-like representation indicating pixels belonging to a region or class

B. A camera cover

C. A model checkpoint only

D. A color profile

Correct Answer: A. An image-like representation indicating pixels belonging to a region or class

Why this is correct: An image-like representation indicating pixels belonging to a region or class directly matches the concept described in the question.

Why B is incorrect: A camera cover does not correctly describe the concept being tested in this question.

Why C is incorrect: A model checkpoint only does not correctly describe the concept being tested in this question.

Why D is incorrect: A color profile does not correctly describe the concept being tested in this question.

Question 57: What is OCR?

A. Optical Character Recognition

B. Object Color Reconstruction

C. Optical Camera Rendering

D. Output Classification Regression

Correct Answer: A. Optical Character Recognition

Why this is correct: Optical Character Recognition directly matches the concept described in the question.

Why B is incorrect: Object Color Reconstruction does not correctly describe the concept being tested in this question.

Why C is incorrect: Optical Camera Rendering does not correctly describe the concept being tested in this question.

Why D is incorrect: Output Classification Regression does not correctly describe the concept being tested in this question.

Question 58: What is the purpose of OCR?

A. Extracting machine-readable text from images or scanned documents

B. Detecting only image edges

C. Generating camera calibration parameters

D. Increasing image sharpness only

Correct Answer: A. Extracting machine-readable text from images or scanned documents

Why this is correct: Extracting machine-readable text from images or scanned documents directly matches the concept described in the question.

Why B is incorrect: Detecting only image edges does not correctly describe the concept being tested in this question.

Why C is incorrect: Generating camera calibration parameters does not correctly describe the concept being tested in this question.

Why D is incorrect: Increasing image sharpness only does not correctly describe the concept being tested in this question.

Question 59: What is face detection?

A. Locating faces in an image or video

B. Determining a person's identity with certainty

C. Changing a face into a 3D model automatically

D. Removing all faces from a dataset

Correct Answer: A. Locating faces in an image or video

Why this is correct: Locating faces in an image or video directly matches the concept described in the question.

Why B is incorrect: Determining a person's identity with certainty does not correctly describe the concept being tested in this question.

Why C is incorrect: Changing a face into a 3D model automatically does not correctly describe the concept being tested in this question.

Why D is incorrect: Removing all faces from a dataset does not correctly describe the concept being tested in this question.

Question 60: What is face recognition?

A. Comparing detected facial features to determine or verify identity

B. Detecting any rectangular region

C. Increasing image resolution

D. Performing only edge detection

Correct Answer: A. Comparing detected facial features to determine or verify identity

Why this is correct: Comparing detected facial features to determine or verify identity directly matches the concept described in the question.

Why B is incorrect: Detecting any rectangular region does not correctly describe the concept being tested in this question.

Why C is incorrect: Increasing image resolution does not correctly describe the concept being tested in this question.

Why D is incorrect: Performing only edge detection does not correctly describe the concept being tested in this question.

Question 61: What is face landmark detection?

A. Detecting predefined key points on a face such as eye or mouth locations

B. Identifying image file types

C. Detecting only full-body objects

D. Compressing a face image

Correct Answer: A. Detecting predefined key points on a face such as eye or mouth locations

Why this is correct: Detecting predefined key points on a face such as eye or mouth locations directly matches the concept described in the question.

Why B is incorrect: Identifying image file types does not correctly describe the concept being tested in this question.

Why C is incorrect: Detecting only full-body objects does not correctly describe the concept being tested in this question.

Why D is incorrect: Compressing a face image does not correctly describe the concept being tested in this question.

Question 62: Which library is widely used for computer vision tasks in Python?

A. OpenCV

B. NumPy only

C. Flask only

D. SQLAlchemy only

Correct Answer: A. OpenCV

Why this is correct: OpenCV directly matches the concept described in the question.

Why B is incorrect: NumPy only does not correctly describe the concept being tested in this question.

Why C is incorrect: Flask only does not correctly describe the concept being tested in this question.

Why D is incorrect: SQLAlchemy only does not correctly describe the concept being tested in this question.

Question 63: What is OpenCV primarily used for?

A. Computer vision and image/video processing

B. Database administration only

C. Web hosting only

D. Spreadsheet editing

Correct Answer: A. Computer vision and image/video processing

Why this is correct: Computer vision and image/video processing directly matches the concept described in the question.

Why B is incorrect: Database administration only does not correctly describe the concept being tested in this question.

Why C is incorrect: Web hosting only does not correctly describe the concept being tested in this question.

Why D is incorrect: Spreadsheet editing does not correctly describe the concept being tested in this question.

Question 64: Which OpenCV function is commonly used to read an image?

A. cv2.imread()

B. cv2.readtext()

C. cv2.openimage()

D. cv2.loadpic()

Correct Answer: A. cv2.imread()

Why this is correct: cv2.imread() directly matches the concept described in the question.

Why B is incorrect: cv2.readtext() does not correctly describe the concept being tested in this question.

Why C is incorrect: cv2.openimage() does not correctly describe the concept being tested in this question.

Why D is incorrect: cv2.loadpic() does not correctly describe the concept being tested in this question.

Question 65: Which OpenCV function is commonly used to display an image in a window?

A. cv2.imshow()

B. cv2.showimagefile()

C. cv2.display()

D. cv2.rendercv()

Correct Answer: A. cv2.imshow()

Why this is correct: cv2.imshow() directly matches the concept described in the question.

Why B is incorrect: cv2.showimagefile() does not correctly describe the concept being tested in this question.

Why C is incorrect: cv2.display() does not correctly describe the concept being tested in this question.

Why D is incorrect: cv2.rendercv() does not correctly describe the concept being tested in this question.

Question 66: Which OpenCV function can convert an image from BGR to grayscale?

A. cv2.cvtColor()

B. cv2.gray()

C. cv2.toGrayImage()

D. cv2.convertGrayOnly()

Correct Answer: A. cv2.cvtColor()

Why this is correct: cv2.cvtColor() directly matches the concept described in the question.

Why B is incorrect: cv2.gray() does not correctly describe the concept being tested in this question.

Why C is incorrect: cv2.toGrayImage() does not correctly describe the concept being tested in this question.

Why D is incorrect: cv2.convertGrayOnly() does not correctly describe the concept being tested in this question.

Question 67: Why does OpenCV often use BGR channel ordering when reading color images?

A. It is the conventional channel ordering used by OpenCV's image-loading functions

B. Because RGB images do not exist

C. Because BGR always has higher quality

D. Because CNNs require BGR only

Correct Answer: A. It is the conventional channel ordering used by OpenCV's image-loading functions

Why this is correct: It is the conventional channel ordering used by OpenCV's image-loading functions directly matches the concept described in the question.

Why B is incorrect: Because RGB images do not exist does not correctly describe the concept being tested in this question.

Why C is incorrect: Because BGR always has higher quality does not correctly describe the concept being tested in this question.

Why D is incorrect: Because CNNs require BGR only does not correctly describe the concept being tested in this question.

Question 68: What is HSV useful for in computer vision?

A. Separating hue, saturation, and value, which can be useful for color-based processing

B. Replacing all CNN layers

C. Storing annotations in SQL

D. Performing OCR automatically

Correct Answer: A. Separating hue, saturation, and value, which can be useful for color-based processing

Why this is correct: Separating hue, saturation, and value, which can be useful for color-based processing directly matches the concept described in the question.

Why B is incorrect: Replacing all CNN layers does not correctly describe the concept being tested in this question.

Why C is incorrect: Storing annotations in SQL does not correctly describe the concept being tested in this question.

Why D is incorrect: Performing OCR automatically does not correctly describe the concept being tested in this question.

Question 69: What does HOG stand for?

A. Histogram of Oriented Gradients

B. High Output Graphics

C. Histogram of Object Groups

D. Hierarchical Optical Geometry

Correct Answer: A. Histogram of Oriented Gradients

Why this is correct: Histogram of Oriented Gradients directly matches the concept described in the question.

Why B is incorrect: High Output Graphics does not correctly describe the concept being tested in this question.

Why C is incorrect: Histogram of Object Groups does not correctly describe the concept being tested in this question.

Why D is incorrect: Hierarchical Optical Geometry does not correctly describe the concept being tested in this question.

Question 70: What is HOG traditionally used for?

A. Describing local shape/edge information for object detection

B. Database indexing

C. Audio enhancement

D. Video compression only

Correct Answer: A. Describing local shape/edge information for object detection

Why this is correct: Describing local shape/edge information for object detection directly matches the concept described in the question.

Why B is incorrect: Database indexing does not correctly describe the concept being tested in this question.

Why C is incorrect: Audio enhancement does not correctly describe the concept being tested in this question.

Why D is incorrect: Video compression only does not correctly describe the concept being tested in this question.

Question 71: What does SIFT stand for?

A. Scale-Invariant Feature Transform

B. Spatial Image Filtering Tool

C. Scale Image Feature Tracker

D. Semantic Inference Feature Transformer

Correct Answer: A. Scale-Invariant Feature Transform

Why this is correct: Scale-Invariant Feature Transform directly matches the concept described in the question.

Why B is incorrect: Spatial Image Filtering Tool does not correctly describe the concept being tested in this question.

Why C is incorrect: Scale Image Feature Tracker does not correctly describe the concept being tested in this question.

Why D is incorrect: Semantic Inference Feature Transformer does not correctly describe the concept being tested in this question.

Question 72: What is a key property of SIFT features?

A. They are designed to be robust to scale and rotation changes

B. They require a neural network

C. They only work on videos

D. They cannot detect local features

Correct Answer: A. They are designed to be robust to scale and rotation changes

Why this is correct: They are designed to be robust to scale and rotation changes directly matches the concept described in the question.

Why B is incorrect: They require a neural network does not correctly describe the concept being tested in this question.

Why C is incorrect: They only work on videos does not correctly describe the concept being tested in this question.

Why D is incorrect: They cannot detect local features does not correctly describe the concept being tested in this question.

Question 73: What is ORB?

A. A fast feature detector and descriptor based on oriented FAST and rotated BRIEF

B. A segmentation neural network

C. A loss function

D. A color space

Correct Answer: A. A fast feature detector and descriptor based on oriented FAST and rotated BRIEF

Why this is correct: A fast feature detector and descriptor based on oriented FAST and rotated BRIEF directly matches the concept described in the question.

Why B is incorrect: A segmentation neural network does not correctly describe the concept being tested in this question.

Why C is incorrect: A loss function does not correctly describe the concept being tested in this question.

Why D is incorrect: A color space does not correctly describe the concept being tested in this question.

Question 74: What is optical flow?

A. The apparent motion of image points between consecutive frames

B. A method for compressing images

C. A static image classification metric

D. A database query language

Correct Answer: A. The apparent motion of image points between consecutive frames

Why this is correct: The apparent motion of image points between consecutive frames directly matches the concept described in the question.

Why B is incorrect: A method for compressing images does not correctly describe the concept being tested in this question.

Why C is incorrect: A static image classification metric does not correctly describe the concept being tested in this question.

Why D is incorrect: A database query language does not correctly describe the concept being tested in this question.

Question 75: What is image registration?

A. Aligning images into a common coordinate system

B. Removing all image pixels

C. Changing image labels

D. Generating random images

Correct Answer: A. Aligning images into a common coordinate system

Why this is correct: Aligning images into a common coordinate system directly matches the concept described in the question.

Why B is incorrect: Removing all image pixels does not correctly describe the concept being tested in this question.

Why C is incorrect: Changing image labels does not correctly describe the concept being tested in this question.

Why D is incorrect: Generating random images does not correctly describe the concept being tested in this question.

Question 76: What is homography commonly used for?

A. Relating points between two views of a planar surface or projective transformations

B. Training a CNN without labels

C. Detecting audio objects

D. Changing image file extensions only

Correct Answer: A. Relating points between two views of a planar surface or projective transformations

Why this is correct: Relating points between two views of a planar surface or projective transformations directly matches the concept described in the question.

Why B is incorrect: Training a CNN without labels does not correctly describe the concept being tested in this question.

Why C is incorrect: Detecting audio objects does not correctly describe the concept being tested in this question.

Why D is incorrect: Changing image file extensions only does not correctly describe the concept being tested in this question.

Question 77: What is camera calibration?

A. Estimating camera parameters that describe how the camera maps 3D points to images

B. Increasing camera megapixels

C. Changing image labels

D. Compressing video

Correct Answer: A. Estimating camera parameters that describe how the camera maps 3D points to images

Why this is correct: Estimating camera parameters that describe how the camera maps 3D points to images directly matches the concept described in the question.

Why B is incorrect: Increasing camera megapixels does not correctly describe the concept being tested in this question.

Why C is incorrect: Changing image labels does not correctly describe the concept being tested in this question.

Why D is incorrect: Compressing video does not correctly describe the concept being tested in this question.

Question 78: What are intrinsic camera parameters?

A. Parameters such as focal length and principal point describing the camera's internal geometry

B. The object's real-world color

C. The number of dataset classes

D. The GPU's clock speed

Correct Answer: A. Parameters such as focal length and principal point describing the camera's internal geometry

Why this is correct: Parameters such as focal length and principal point describing the camera's internal geometry directly matches the concept described in the question.

Why B is incorrect: The object's real-world color does not correctly describe the concept being tested in this question.

Why C is incorrect: The number of dataset classes does not correctly describe the concept being tested in this question.

Why D is incorrect: The GPU's clock speed does not correctly describe the concept being tested in this question.

Question 79: What are extrinsic camera parameters?

A. Camera pose parameters describing position and orientation relative to a scene

B. Pixel intensity values

C. Image compression settings

D. Neural network weights only

Correct Answer: A. Camera pose parameters describing position and orientation relative to a scene

Why this is correct: Camera pose parameters describing position and orientation relative to a scene directly matches the concept described in the question.

Why B is incorrect: Pixel intensity values does not correctly describe the concept being tested in this question.

Why C is incorrect: Image compression settings does not correctly describe the concept being tested in this question.

Why D is incorrect: Neural network weights only does not correctly describe the concept being tested in this question.

Question 80: What is depth estimation?

A. Estimating the distance of scene points from the camera

B. Estimating image file size

C. Predicting class labels only

D. Detecting image edges only

Correct Answer: A. Estimating the distance of scene points from the camera

Why this is correct: Estimating the distance of scene points from the camera directly matches the concept described in the question.

Why B is incorrect: Estimating image file size does not correctly describe the concept being tested in this question.

Why C is incorrect: Predicting class labels only does not correctly describe the concept being tested in this question.

Why D is incorrect: Detecting image edges only does not correctly describe the concept being tested in this question.

Question 81: What is stereo vision?

A. Estimating depth using images from two or more viewpoints

B. Using two CNN optimizers

C. Using two labels per image

D. Converting two images to one JPEG

Correct Answer: A. Estimating depth using images from two or more viewpoints

Why this is correct: Estimating depth using images from two or more viewpoints directly matches the concept described in the question.

Why B is incorrect: Using two CNN optimizers does not correctly describe the concept being tested in this question.

Why C is incorrect: Using two labels per image does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting two images to one JPEG does not correctly describe the concept being tested in this question.

Question 82: What is a disparity map in stereo vision?

A. A map representing the difference in corresponding pixel positions between stereo images

B. A classification confusion matrix

C. A list of camera names

D. A segmentation label dictionary

Correct Answer: A. A map representing the difference in corresponding pixel positions between stereo images

Why this is correct: A map representing the difference in corresponding pixel positions between stereo images directly matches the concept described in the question.

Why B is incorrect: A classification confusion matrix does not correctly describe the concept being tested in this question.

Why C is incorrect: A list of camera names does not correctly describe the concept being tested in this question.

Why D is incorrect: A segmentation label dictionary does not correctly describe the concept being tested in this question.

Question 83: What is 3D reconstruction?

A. Recovering 3D structure or geometry of a scene from visual observations

B. Converting RGB to grayscale

C. Training a 2D classifier only

D. Removing image backgrounds

Correct Answer: A. Recovering 3D structure or geometry of a scene from visual observations

Why this is correct: Recovering 3D structure or geometry of a scene from visual observations directly matches the concept described in the question.

Why B is incorrect: Converting RGB to grayscale does not correctly describe the concept being tested in this question.

Why C is incorrect: Training a 2D classifier only does not correctly describe the concept being tested in this question.

Why D is incorrect: Removing image backgrounds does not correctly describe the concept being tested in this question.

Question 84: What is a generative adversarial network (GAN)?

A. A system with a generator and discriminator trained in an adversarial setup

B. A classical edge detector

C. A database engine

D. A camera calibration method

Correct Answer: A. A system with a generator and discriminator trained in an adversarial setup

Why this is correct: A system with a generator and discriminator trained in an adversarial setup directly matches the concept described in the question.

Why B is incorrect: A classical edge detector does not correctly describe the concept being tested in this question.

Why C is incorrect: A database engine does not correctly describe the concept being tested in this question.

Why D is incorrect: A camera calibration method does not correctly describe the concept being tested in this question.

Question 85: What does the discriminator do in a GAN?

A. It tries to distinguish real data from generated data

B. It generates all images without feedback

C. It detects only edges

D. It calculates camera focal length

Correct Answer: A. It tries to distinguish real data from generated data

Why this is correct: It tries to distinguish real data from generated data directly matches the concept described in the question.

Why B is incorrect: It generates all images without feedback does not correctly describe the concept being tested in this question.

Why C is incorrect: It detects only edges does not correctly describe the concept being tested in this question.

Why D is incorrect: It calculates camera focal length does not correctly describe the concept being tested in this question.

Question 86: What does the generator do in a GAN?

A. It generates synthetic samples intended to resemble real data

B. It labels all training images manually

C. It performs NMS

D. It calibrates the camera

Correct Answer: A. It generates synthetic samples intended to resemble real data

Why this is correct: It generates synthetic samples intended to resemble real data directly matches the concept described in the question.

Why B is incorrect: It labels all training images manually does not correctly describe the concept being tested in this question.

Why C is incorrect: It performs NMS does not correctly describe the concept being tested in this question.

Why D is incorrect: It calibrates the camera does not correctly describe the concept being tested in this question.

Question 87: What is a Vision Transformer (ViT)?

A. A transformer architecture that processes images using image patches/tokens

B. A type of image file

C. A camera sensor

D. A traditional edge detector

Correct Answer: A. A transformer architecture that processes images using image patches/tokens

Why this is correct: A transformer architecture that processes images using image patches/tokens directly matches the concept described in the question.

Why B is incorrect: A type of image file does not correctly describe the concept being tested in this question.

Why C is incorrect: A camera sensor does not correctly describe the concept being tested in this question.

Why D is incorrect: A traditional edge detector does not correctly describe the concept being tested in this question.

Question 88: How are images commonly represented for a Vision Transformer?

A. As sequences of patch embeddings or visual tokens

B. As SQL rows only

C. As audio waveforms

D. As bounding boxes only

Correct Answer: A. As sequences of patch embeddings or visual tokens

Why this is correct: As sequences of patch embeddings or visual tokens directly matches the concept described in the question.

Why B is incorrect: As SQL rows only does not correctly describe the concept being tested in this question.

Why C is incorrect: As audio waveforms does not correctly describe the concept being tested in this question.

Why D is incorrect: As bounding boxes only does not correctly describe the concept being tested in this question.

Question 89: What is multimodal computer vision?

A. Working with visual data together with other modalities such as text or audio

B. Using multiple image sizes only

C. Using multiple GPUs only

D. Using multiple file formats only

Correct Answer: A. Working with visual data together with other modalities such as text or audio

Why this is correct: Working with visual data together with other modalities such as text or audio directly matches the concept described in the question.

Why B is incorrect: Using multiple image sizes only does not correctly describe the concept being tested in this question.

Why C is incorrect: Using multiple GPUs only does not correctly describe the concept being tested in this question.

Why D is incorrect: Using multiple file formats only does not correctly describe the concept being tested in this question.

Question 90: What is visual question answering (VQA)?

A. Answering natural-language questions about an image

B. Detecting image file corruption

C. Generating bounding boxes only

D. Converting video to grayscale

Correct Answer: A. Answering natural-language questions about an image

Why this is correct: Answering natural-language questions about an image directly matches the concept described in the question.

Why B is incorrect: Detecting image file corruption does not correctly describe the concept being tested in this question.

Why C is incorrect: Generating bounding boxes only does not correctly describe the concept being tested in this question.

Why D is incorrect: Converting video to grayscale does not correctly describe the concept being tested in this question.

Question 91: What is image captioning?

A. Generating a natural-language description of an image

B. Assigning a single numeric pixel value

C. Detecting camera parameters

D. Removing all objects

Correct Answer: A. Generating a natural-language description of an image

Why this is correct: Generating a natural-language description of an image directly matches the concept described in the question.

Why B is incorrect: Assigning a single numeric pixel value does not correctly describe the concept being tested in this question.

Why C is incorrect: Detecting camera parameters does not correctly describe the concept being tested in this question.

Why D is incorrect: Removing all objects does not correctly describe the concept being tested in this question.

Question 92: What is a common loss for multi-class image classification?

A. Cross-entropy loss

B. IoU only

C. Hough loss

D. JPEG loss

Correct Answer: A. Cross-entropy loss

Why this is correct: Cross-entropy loss directly matches the concept described in the question.

Why B is incorrect: IoU only does not correctly describe the concept being tested in this question.

Why C is incorrect: Hough loss does not correctly describe the concept being tested in this question.

Why D is incorrect: JPEG loss does not correctly describe the concept being tested in this question.

Question 93: What is precision in object detection?

A. The fraction of predicted positive detections that are correct

B. The fraction of all negatives detected

C. The number of pixels per image

D. The model's image resolution

Correct Answer: A. The fraction of predicted positive detections that are correct

Why this is correct: The fraction of predicted positive detections that are correct directly matches the concept described in the question.

Why B is incorrect: The fraction of all negatives detected does not correctly describe the concept being tested in this question.

Why C is incorrect: The number of pixels per image does not correctly describe the concept being tested in this question.

Why D is incorrect: The model's image resolution does not correctly describe the concept being tested in this question.

Question 94: What is recall in object detection?

A. The fraction of actual positive objects that are successfully detected

B. The number of false positives divided by all pixels

C. The size of the input image

D. The number of model layers

Correct Answer: A. The fraction of actual positive objects that are successfully detected

Why this is correct: The fraction of actual positive objects that are successfully detected directly matches the concept described in the question.

Why B is incorrect: The number of false positives divided by all pixels does not correctly describe the concept being tested in this question.

Why C is incorrect: The size of the input image does not correctly describe the concept being tested in this question.

Why D is incorrect: The number of model layers does not correctly describe the concept being tested in this question.

Question 95: What is mean Average Precision (mAP)?

A. A commonly used aggregate metric for object detection performance across classes and/or IoU thresholds

B. A method for image resizing

C. A color space

D. A convolution operation

Correct Answer: A. A commonly used aggregate metric for object detection performance across classes and/or IoU thresholds

Why this is correct: A commonly used aggregate metric for object detection performance across classes and/or IoU thresholds directly matches the concept described in the question.

Why B is incorrect: A method for image resizing does not correctly describe the concept being tested in this question.

Why C is incorrect: A color space does not correctly describe the concept being tested in this question.

Why D is incorrect: A convolution operation does not correctly describe the concept being tested in this question.

Question 96: What is overfitting in a computer vision model?

A. When a model learns training data too specifically and generalizes poorly

B. When an image is too large

C. When a camera has low resolution

D. When every prediction is correct

Correct Answer: A. When a model learns training data too specifically and generalizes poorly

Why this is correct: When a model learns training data too specifically and generalizes poorly directly matches the concept described in the question.

Why B is incorrect: When an image is too large does not correctly describe the concept being tested in this question.

Why C is incorrect: When a camera has low resolution does not correctly describe the concept being tested in this question.

Why D is incorrect: When every prediction is correct does not correctly describe the concept being tested in this question.

Question 97: Which technique can help reduce overfitting in vision models?

A. Data augmentation

B. Deleting the validation set

C. Increasing label noise

D. Using the test set for training

Correct Answer: A. Data augmentation

Why this is correct: Data augmentation directly matches the concept described in the question.

Why B is incorrect: Deleting the validation set does not correctly describe the concept being tested in this question.

Why C is incorrect: Increasing label noise does not correctly describe the concept being tested in this question.

Why D is incorrect: Using the test set for training does not correctly describe the concept being tested in this question.

Question 98: Why must the test set generally remain separate from training?

A. To provide an unbiased estimate of performance on unseen data

B. To increase image brightness

C. To reduce the number of classes

D. To make convolution faster

Correct Answer: A. To provide an unbiased estimate of performance on unseen data

Why this is correct: To provide an unbiased estimate of performance on unseen data directly matches the concept described in the question.

Why B is incorrect: To increase image brightness does not correctly describe the concept being tested in this question.

Why C is incorrect: To reduce the number of classes does not correctly describe the concept being tested in this question.

Why D is incorrect: To make convolution faster does not correctly describe the concept being tested in this question.

Question 99: What is data leakage in computer vision?

A. When information from evaluation data or future information improperly influences training

B. When an image contains a bright region

C. When a camera lens is dirty

D. When a model uses convolution

Correct Answer: A. When information from evaluation data or future information improperly influences training

Why this is correct: When information from evaluation data or future information improperly influences training directly matches the concept described in the question.

Why B is incorrect: When an image contains a bright region does not correctly describe the concept being tested in this question.

Why C is incorrect: When a camera lens is dirty does not correctly describe the concept being tested in this question.

Why D is incorrect: When a model uses convolution does not correctly describe the concept being tested in this question.

Question 100: Why can duplicate or near-duplicate images across train and test be problematic?

A. They can make test performance look artificially high

B. They always reduce accuracy

C. They remove all labels

D. They prevent image loading

Correct Answer: A. They can make test performance look artificially high

Why this is correct: They can make test performance look artificially high directly matches the concept described in the question.

Why B is incorrect: They always reduce accuracy does not correctly describe the concept being tested in this question.

Why C is incorrect: They remove all labels does not correctly describe the concept being tested in this question.

Why D is incorrect: They prevent image loading does not correctly describe the concept being tested in this question.

Final Answer Key

1. A 2. A 3. A 4. A 5. A 6. A 7. A 8. A 9. A 10. A 11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. A 19. A 20. A 21. A 22. A 23. A 24. A 25. A

26. A 27. A 28. A 29. A 30. A 31. A 32. A 33. A 34. A 35. A 36. A 37. A 38. A 39. A 40. A 41. A 42. A 43. A 44. A 45. A 46. A 47. A 48. A 49. A 50. A

51. A 52. A 53. A 54. A 55. A 56. A 57. A 58. A 59. A 60. A 61. A 62. A 63. A 64. A 65. A 66. A 67. A 68. A 69. A 70. A 71. A 72. A 73. A 74. A 75. A

76. A 77. A 78. A 79. A 80. A 81. A 82. A 83. A 84. A 85. A 86. A 87. A 88. A 89. A 90. A 91. A 92. A 93. A 94. A 95. A 96. A 97. A 98. A 99. A 100. A